Library Management System

1) Introduction

1.1 Purpose

This document describes the requirements of a library Management System and it's main functions.

1.2 - Scope

The system manages books, members, input/output issue/return activities, fines and library records.

1.3 Overview

this system helps students and librarians and adm~~inistr~~ manage library operations easily. It keeps track of books, members and transactions

2) General Description

The system is used by students, librarians and administrators. It provides features such as:

- Book management
- Member management
- Book issue and return
- Fine calculation
- Search and reports.

3) Functional Requirements

3.1 Book Management

→ Add, update and remove book details.
→ Check book availability.

2) Members management -
- Register and manage member details.
- Maintain member borrowing records

3.3 - Issue and Return
- Issue available ~~to~~ books to members
- Record returned books.
- Update book availability.

3.4 → Fine Management -
⇒ Calculate finds for late returns.
→ Maintain fine payment records

3.5 → Search and Report
- Search books by title, author or category
- Generate basic library reports.

4. Non-functional Requirements

4.1 Availability -
The system should be available during library working hours.

4.2 - Performance
Searches and transactions should be completed quickly.

4.3. Security
Only authorised users should access

administrative functions.

a 4 -) Reliability. ~~Performance~~. Reliability
Book and member records should be stored ~~completed~~ ~~quickly~~. accurately

4. 5

5) Design Constraints

- The system requires a database to store records.
- Only authorized staff can modify library.
- A reliable network may be required for online access.

6) Hardware and Software requirements

Hardware: computer/server, barcode scanner, network connection

Software: Operating system, database, web/application server and web browser

7) Preliminary ~~budget~~ schedule & Budget
6 months with budget of $37000 the bar project including development, hardware, software and maintenance costs.

division of cost ?

7/10

Preliminary Schedule

Activity	Duration
Requirement Analysis	1 week
System Design	1 week
Development	2 weeks
Testing	1 week
Deployment & Documentation	1 week
Total	6 weeks

Preliminary Budget

Resource	Estimated cost
Development	₹20,000
Testing	₹5000
Hardware / Server	₹5000
Software / Tools	₹3000
Maintenance	₹4000
Total	₹37000